■ ENERGY CONSUMPTION FORECASTING SYSTEM

Using Machine Learning

Artificial Intelligence	Machine Learning	Energy Analytics	Smart Energy Management

PROJECT TEAM

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TABLE OF CONTENTS

1. INTRODUCTION

Energy consumption forecasting is the process of predicting future electricity usage based on environmental and operational conditions. In modern smart systems, energy demand continuously changes depending on weather conditions, occupancy levels, HVAC usage, lighting usage, renewable energy generation, and time patterns.

Traditional manual estimation methods are inaccurate because energy usage is dynamic and nonlinear in nature. To solve this problem, an AI-powered Energy Consumption Forecasting System was developed, capable of accurately predicting building energy usage using Machine Learning techniques. The system analyzes temperature, humidity, occupancy, appliance usage, renewable energy, and time-based patterns to predict future energy consumption and electricity bills.

Machine Learning Type & Algorithm

The project uses **Supervised Machine Learning** because the dataset contains input features and labeled output values. The model learns the relationship between Input Features and Energy Consumption. The primary algorithm is the **Gradient Boosting Regressor** — an ensemble method that combines multiple decision trees. It was selected because it handles nonlinear relationships effectively, provides high accuracy on structured datasets, and reduces prediction errors iteratively.

Technologies Used

Component	Technology
Programming Language	Python
Machine Learning	Scikit-learn
Frontend	React.js
Backend	FastAPI
Deployment	Render & Vercel

2. PROBLEM STATEMENT

Existing Problem

Energy consumption changes continuously due to environmental conditions, occupancy behavior, HVAC usage, lighting demand, renewable energy availability, and seasonal and time-based variations. Traditional estimation methods are manual, inefficient, and inaccurate — unable to capture complex nonlinear relationships. As a result, electricity wastage increases, energy planning becomes difficult, and cost optimization becomes challenging.

Need for the System

Modern buildings and smart systems require intelligent solutions capable of predicting future energy demand, estimating electricity bills, optimizing energy usage, and reducing energy wastage. This creates the need for an AI-based forecasting system.

Proposed Solution

The proposed Energy Consumption Forecasting System uses Machine Learning techniques to predict energy usage automatically. The system collects environmental and operational inputs, preprocesses the data, engineers advanced features, applies Machine Learning algorithms, predicts energy consumption, and estimates electricity bills — providing accurate and intelligent energy forecasting.

Importance & Role of Machine Learning

The project supports smart homes, commercial buildings, smart city infrastructure, and sustainable energy systems. It helps improve energy efficiency, reduce electricity wastage, optimize appliance usage, lower operational costs, and support intelligent decision-making. Machine Learning solves the problem by learning hidden patterns from data, identifying nonlinear relationships, automatically improving prediction accuracy, and adapting to varying energy behaviors — learning directly from historical patterns rather than manually defined rules.

3. OBJECTIVES

The major objectives of the project are:

1. Generate a Realistic Energy Dataset
Create a synthetic but realistic dataset that simulates real-world building energy behavior.

2. Analyze Energy Consumption Patterns
Study how different environmental and operational factors affect electricity usage.

3. Preprocess and Clean Data
Prepare the dataset by encoding categorical values, extracting time-based features, and transforming data into a usable format.

4. Perform Feature Engineering
Create advanced engineered features such as cyclical time encodings, interaction features, and occupancy-energy relationships to improve prediction accuracy.

5. Apply Machine Learning Algorithms
Train and evaluate Machine Learning models capable of forecasting energy consumption accurately.

6. Predict Future Energy Consumption
Forecast hourly, daily, and monthly electricity usage based on user inputs.

7. Estimate Electricity Bills
Calculate estimated electricity costs using predicted energy usage.

8. Build a Full-Stack Application
Integrate React frontend, FastAPI backend, and Machine Learning model into a complete intelligent forecasting system.

9. Deploy the System Online
Deploy the frontend and backend using cloud platforms for real-time public accessibility.

4. DATASET DESCRIPTION

A dataset is a collection of structured data used to train and test Machine Learning models. In this project, a synthetic but realistic dataset containing **8,000 hourly records** was generated to simulate real-world building energy consumption behavior. The dataset was created using realistic energy equations so that the model learns genuine energy patterns instead of random values.

Why a Synthetic Dataset Was Used

Real-world datasets often contain missing values, require authentication, have inconsistent records, and include privacy restrictions. Therefore, a physics-based synthetic dataset was generated programmatically to simulate weather variation, occupancy behavior, HVAC usage, lighting patterns, renewable energy generation, and peak-hour demand — providing clean, realistic, and reliable training data.

Dataset Size

Attribute	Value
Total Records	8,000
Data Type	Hourly Time-Series Data
Original Features	11
Engineered Features	27+

Feature Description

Feature	Description
Timestamp	Date and time of the record
Temperature	Environmental temperature
Humidity	Atmospheric humidity
SquareFootage	Building size
Occupancy	Number of occupants
HVACUsage	AC/heating usage (On/Off)
LightingUsage	Lighting status (On/Off)
RenewableEnergy	Solar energy generation
DayOfWeek	Day information
Holiday	Holiday indicator (Yes/No)
EnergyConsumption	Target variable (kWh)

Energy Consumption Formula

EnergyConsumption *= (Base Load + HVAC Load + Occupancy Load + Lighting Load – Renewable Energy Offset) × Peak Hour Factor + Noise*

5. DATA ANALYSIS & PREPROCESSING

Data preprocessing converts raw data into a structured format suitable for Machine Learning. The preprocessing stage included timestamp processing, categorical encoding, feature engineering, and cyclical encoding. The dataset was analyzed using statistical summaries, feature distributions, and datatype inspection.

Key Observations

- Energy usage increases during evening peak hours
- HVAC demand rises during extreme temperatures
- Higher occupancy increases electricity consumption
- Renewable energy reduces daytime electricity demand

Missing Value Handling

The dataset was generated programmatically, therefore no missing values were present.

Timestamp Processing

The timestamp column was converted into Hour, Day, Month, and DayOfWeek features. These features help the model understand time-based energy patterns.

Categorical Encoding

Categorical values were converted into numerical form to make them usable by the Machine Learning model:

Original Value	Encoded Value
On	1
Off	0
Yes	1
No	0

Feature Engineering

Feature Engineering was one of the most important stages of the project. The final system uses **27+ engineered features** to improve prediction accuracy.

Cyclical Encoding

Time is cyclical in nature — Hour 23 and Hour 0 are close, December and January are adjacent. To preserve this continuity, sine and cosine transformations were applied:

$$\textbf{HourSin} = \sin(2\pi \times Hour / 24) \qquad \textbf{HourCos} = \cos(2\pi \times Hour / 24)$$

Interaction Features

Interaction features capture relationships between multiple variables and significantly improved model performance and forecasting accuracy:

Interaction Feature	Purpose
Temperature × Occupancy	High heat combined with crowd increases AC demand
Temperature × HVAC	Weather directly affects cooling usage

Occupancy × Lighting	More occupants increase lighting demand
Renewable Ratio	Measures solar energy generation efficiency

6. ML METHODOLOGY

ML methodology refers to the process of selecting, training, and evaluating the Machine Learning model used for prediction. This project uses **Supervised Regression Learning** because the target variable EnergyConsumption is continuous numerical data.

Algorithm — Gradient Boosting Regressor

The main algorithm is the Gradient Boosting Regressor, selected because it provides high accuracy, handles nonlinear relationships, works well on structured datasets, and improves prediction iteratively. Gradient Boosting builds multiple decision trees sequentially — each new tree learns from previous errors, corrects mistakes, and improves prediction accuracy — resulting in strong forecasting performance.

Model Training Process

The training process included: data preprocessing, feature engineering, train-test split, model training, hyperparameter tuning, and model evaluation.

Train-Test Split

Split	Percentage	Purpose
Training Data	80%	Helps the model learn patterns from data
Testing Data	20%	Evaluates model performance on unseen data

Hyperparameter Tuning & Final Parameters

Different parameters were tested including learning rate, number of estimators, tree depth, and subsampling to improve prediction accuracy. The following final values produced the best model performance:

Parameter	Value	Effect
n_estimators	500	Number of boosting rounds
learning_rate	0.05	Step size for each tree update
max_depth	5	Maximum depth of each tree
subsample	0.85	Fraction of data used per tree

7. RESULTS & EVALUATION

Model evaluation measures how accurately the Machine Learning model predicts energy consumption. Multiple standard regression metrics were used to assess the final model.

Final Model Performance

0.966

R^2 Score — Model Accuracy

96.6% of energy consumption variation was successfully predicted by the model. This indicates very high prediction accuracy and confirms the model has effectively learned the energy patterns from the dataset.

Evaluation Metrics

Metric	Purpose	Achieved
R^2 Score	Measures overall prediction accuracy	0.966 (96.6%)
RMSE	Root Mean Square — penalises large errors	Low
MAE	Mean Absolute Error — average absolute deviation	Low

8. USE CASE

Applications

This system can be used in smart homes, office buildings, commercial infrastructures, and smart city energy systems. The project helps monitor and forecast electricity usage efficiently across a wide range of real-world scenarios.

Benefits

- Estimate electricity bills accurately before they arrive
- Reduce energy wastage through informed consumption decisions
- Optimize appliance usage based on forecasted demand
- Improve overall energy efficiency across buildings
- Understand detailed energy consumption patterns over time
- Support better energy planning and sustainable energy management

9. TOOLS USED

Component	Technology
Programming Language	Python
ML Libraries	Scikit-learn, Pandas, NumPy
Frontend	React.js
Backend	FastAPI
Deployment	Render & Vercel
Version Control	GitHub

Library Details

Library / Framework	Role & Usage

Pandas	Data preprocessing, data analysis, and dataset handling
NumPy	Numerical computations and mathematical operations
Scikit-learn	Model training, evaluation, and hyperparameter tuning
React.js	Building the interactive user interface and dashboard
FastAPI	Creating APIs and connecting the ML model with the frontend

10. CONCLUSION & FUTURE SCOPE

Conclusion

The Energy Consumption Forecasting System successfully combines Machine Learning, Feature Engineering, Full-Stack Development, and Cloud Deployment to build an intelligent energy prediction platform. The model achieved an **R^2 Score of 0.966**, which indicates very high prediction accuracy. The system can accurately predict energy consumption, daily and monthly usage, and electricity bills using environmental and operational features. This project demonstrates how Machine Learning can be effectively applied to solve real-world energy management problems.

Future Scope

The following enhancements can further improve accuracy and real-world applicability:

Enhancement	Description
Real IoT Smart-Meter Integration	Connect to live smart meters for real-time data collection and live forecasting
Deep Learning — LSTM Models	Use Long Short-Term Memory networks for sequential time-series energy forecasting
Live Weather API Integration	Fetch real-time weather data to dynamically improve prediction inputs
Mobile Application Development	Build iOS and Android apps to make the system accessible to end users
Smart Energy-Saving Recommendations	AI-powered suggestions to help users actively reduce electricity consumption
Advanced Analytics Dashboard	Enhanced visualizations with detailed energy reports and trend analysis

End of Documentation | A. Mohan · G. Nimitha · K. Sindhuja · K. Srikruthika